

CCNA 200-301

CERTIFICATE NUMBER : CERT20240310525



This is to certify that :

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Has completed a 32 hours program of Cisco CCNA Training at the IDN.ID Training Center, Jakarta, on 4, 5, 6, 7 Maret 2024
The student was able to obtain proficiency :

1. Network Fundamentals

- Explain the role and function of network components
- Describe characteristics of network topology architectures
- Compare physical interface and cabling types
- Compare TCP to UDP
- Configure and verify IPv4 addressing and subnetting
- Describe the need for private IPv4 addressing
- Configure and verify IPv6 addressing and prefix
- Compare IPv6 address types
 - a) Global unicast
 - b) Unique local
 - c) Link local
 - d) Anycast
 - e) Multicast
 - f) Modified EUI 64
- Describe wireless principles
 - a) 1.11.a Nonoverlapping Wi-Fi channels
 - b) 1.11.b SSID
 - c) 1.11.c RF
 - d) 1.11.d Encryption
- Explain virtualization fundamentals (virtual machines)
- Describe switching concepts
 - a) 1.13.a MAC learning and aging
 - b) 1.13.b Frame switching
 - c) 1.13.c Frame flooding
 - d) 1.13.d MAC address table

2. Network Access

- Configure and verify VLANs (normal range) spanning multiple switches
 - a) 2.1.a Access ports (data and voice)
 - b) 2.1.b Default VLAN
 - c) 2.1.c Connectivity 2.2 Configure and verify interswitch connectivity
 - a) 2.2.a Trunk ports
 - b) 2.2.b 802.1Q
 - c) 2.2.c Native VLAN
 - d) 2.3 Configure and verify Layer 2 discovery protocols (Cisco Discovery Protocol and LLDP)
 - e) 2.4 Configure and verify (Layer 2/Layer 3) EtherChannel (LACP)
 - f) 2.5 Describe the need for and basic operations of Rapid PVST+ Spanning Tree Protocol and identify basic operations
 - a) 2.5.a Root port, root bridge (primary/secondary), and other port names
 - b) 2.5.b Port states (forwarding/blocking)
 - c) 2.5.c PortFast benefits
 - g) 2.6 Compare Cisco Wireless Architectures and AP modes
 - h) 2.9 Configure the components of a wireless LAN access for client connectivity using GUI only such as WLAN creation, security settings, QoS profiles, and advanced WLAN settings

3. IP Connectivity

- 3.1 Interpret the components of routing table
 - a) 3.1.a Routing protocol code
 - b) 3.1.b Prefix
 - c) 3.1.c Network mask
 - d) 3.1.d Next hop
 - e) 3.1.e Administrative distance
 - f) 3.1.f Metric
 - g) 3.1.g Gateway of last resort
- 3.2 Determine how a router makes a forwarding decision by default
 - a) 3.2.a Longest match
 - b) 3.2.b Administrative distance
 - c) 3.2.c Routing protocol metric
- 3.3 Configure and verify IPv4 and IPv6 static routing
 - a) 3.3.a Default route
 - b) 3.3.b Network route
 - c) 3.3.c Host route
 - d) 3.3.d Floating static
- 3.4 Configure and verify single area OSPFv2
 - a) 3.4.a Neighbor adjacencies
 - b) 3.4.b Point-to-point
 - c) 3.4.c Broadcast (DR/BDR selection)
 - d) 3.4.d Router ID
- 3.5 Describe the purpose of first hop redundancy protocol

4. IP Services

- 4.1 Configure and verify inside source NAT using static and pools
- 4.2 Configure and verify NTP operating in a client and server mode
- 4.3 Explain the role of DHCP and DNS within the network
- 4.4 Explain the function of SNMP in network operations including facilities and levels
- 4.5 Describe the use of syslog features including facilities and levels
- 4.6 Configure and verify DHCP client and relay
- 4.7 Explain the forwarding per-hop behavior (PHB) for QoS such as classification, marking, queuing, congestion, policing, shaping
- 4.8 Configure network devices for remote access using SSH
- 4.9 Describe the capabilities and function of TFTP/FTP in the network

5. Security Fundamentals

- Configure device access control using local passwords
- Describe remote access and site-to-site VPNs
- Configure and verify access control lists
- Configure Layer 2 security features (DHCP snooping, dynamic ARP inspection, and port security)
- Differentiate authentication, authorization, and accounting concepts
- Describe wireless security protocols (WPA, WPA2, and WPA3)
- Configure WLAN using WPA2 PSK using the GUI

6. Automation and Programmability

- Explain how automation impacts network management
- Compare traditional networks with controller-based networking
- Describe controller-based and software defined architectures
 - a) 6.3.a Separation of control plane and data plane
 - b) 6.3.b North-bound and south-bound APIs
 - c) Compare traditional campus device management with Cisco DNA Center enabled devicemanagement
- Recognize the capabilities of configuration management mechanisms Puppet, Chef, and Ansible
- Interpret JSON encoded data

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